

- with other members of the grid.
- **Commuting incentives:**
Government departments can know which citizens use their cars daily, and offer them discounts and advantages to encourage them to take public transport.
- **Improved waste management:**
Blockchain can provide the public and waste collection services with real-time information on containers so that they can always know whether these are full or empty.
- **Better public participation:**
Blockchain platforms guarantee the security, reliability, transparency and anonymity of public consultations such as elections, surveys, referendums, etc. In addition, blockchain technology can benefit other services such as water resource management, air quality control, or park and garden care.

OrgBook architecture

The blockchain platform, with its decentralised network based architecture, is commonly used in fintech industries like banking, insurance and capital markets for a lot of use cases like know your customer (KYC), credit risk management, and fee calculations or draft approvals, to name a few. For some years now, there are many use cases being experimented in other domains too, like the healthcare

industry for distributed patient and doctor database handling using a secured public network, or the retail industry for processing inventories.

The Blockchain LIVE conference of 2019 had some excellent discussions on how to use blockchain to invent options for next generation infrastructure when designing smart cities. One of the key topics for discussion was how blockchain and smart city initiatives marry in terms of solutions and feature provisions. Transparency and security are core fundamentals in smart city planning and infrastructure requirement. There are many use cases in smart city initiatives that can use cloud based blockchain platforms for infrastructure development and management.

In British Columbia, a community effort called the Verifiable Organizations Network (VON) has put together the OrgBook, a searchable public directory of legally registered organisations based on the Hyperledger platform. OrgBook showcases the benefits of an open, unified trusted network of various organisations and associations for the benefit of people and services across a city or province.

The implementation of OrgBook in smart city administration using blockchain based distributed network administration really helps in cutting red tape (reducing paper work and delays in administrative matters),

and improves the efficiency of public administration with real-time workflow management like social security verification, and address registration/de-registration, to name a few.

In blockchain based architecture like that of OrgBook, smart city administration is handled efficiently by designing various use cases as decentralised solutions (service fabric implementation). The decentralised database is used to display public data in a verifiable and authentic manner, empowering individuals (smart city members or the government authority) or entities (different role based access in administration services) to manage their identity (without the need of manual verifiable services).

The architecture involves the development of a mobility service called Citizen Wallet App, which has the following features:

- End users and associated family members to identify, store and share DID (decentralised identifiers) based proofs
- Service for associations and member groups to issue/verify DID based proofs
- Administrative functions like authorisation of certificates, certificate issuance and public notices
- Verification services like authorising identity, providing access rights and approvals

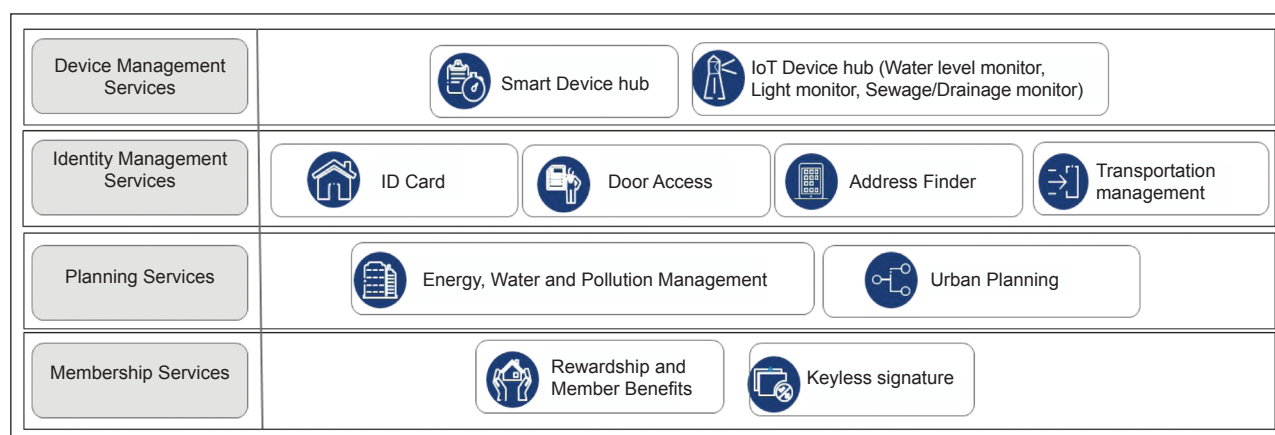


Figure 2: Smart city use cases in Hyperledger OrgBook